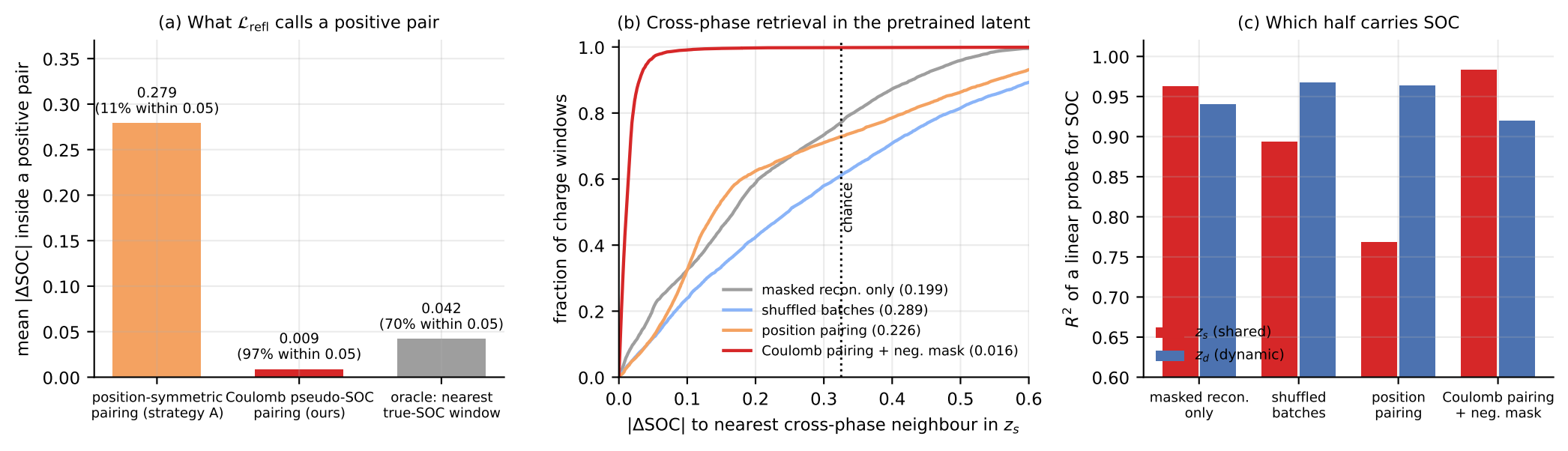
(a) What $\mathcal{L}_{\text{refl}}$ calls a positive pair
mean |ΔSOC| inside a positive pair
0.279
(11% within 0.05)
0.009
(97% within 0.05)
0.042
(70% within 0.05)
position-symmetric pairing (strategy A)
Coulomb pseudo-SOC pairing (ours)
oracle: nearest true-SOC window
(b) Cross-phase retrieval in the pretrained latent
fraction of charge windows
chance
masked recon. only (0.199)
shuffled batches (0.289)
position pairing (0.226)
Coulomb pairing + neg. mask (0.016)
|ΔSOC| to nearest cross-phase neighbour in z_s
(c) Which half carries SOC
R^2 of a linear probe for SOC
z_s (shared)
z_d (dynamic)
masked recon. only
shuffled batches
position pairing
Coulomb pairing + neg. mask